RESEARCH PROGRAM BRIEF

Nanomaterials, battery interfaces and phase-field simulation

How our paper connects to the program

What our paper contributes

The paper studies contact between lithium metal and the solid electrolyte in a battery. Small gaps can reduce the area carrying current and increase electrical resistance. We examine how the cathode's pressure history and the contact state at the start of discharge contribute to that resistance, then propose measurements to help separate their effects.

Why this relates to nanomaterials

Nanostructured coatings, composite interlayers and nanoscale defects can affect how lithium deposits and how well the interface stays connected. A published silver-carbon nanocomposite interlayer, for example, regulates lithium deposition [1]. Our paper addresses the contact and stress problems that these materials aim to improve.

How phase-field simulations can help

Phase-field simulations follow how material regions and their boundaries change over time. Combined with ion transport and mechanics, they can represent growing voids and lithium deposits, and show where current or stress concentrates. Published work has applied this approach to interfacial nanovoids in solid-state batteries [2].

- Compare thin coatings and interlayers with different material properties.
- Study how pores, grain boundaries and small defects affect interface stability.
- Identify promising structures and informative conditions for experimental testing.

Value to the program

This connects nanomaterial design to measurable battery behavior. Simulations can help prioritize what to fabricate and test, while experiments provide the material inputs and checks needed for reliable predictions.

Scope: Our paper uses reduced contact and spatial continuum models. Direct nanoscale phase-field modeling is a possible extension, with suitable material data; it does not track individual atoms.

[1] Lee et al., Nature Energy (2020): silver-carbon nanocomposite anodes. doi:10.1038/s41560-020-0575-z

[2] Kamikawa et al., Communications Materials (2024): phase-field modeling of nanodefects and nanovoids. doi:10.1038/s43246-024-00600-6